

Research Statement

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Research Vision and Alignment with KFUPM

My research program is centered on **trustworthy artificial intelligence**, **secure machine learning**, and **hybrid quantum-classical computing** for safety-critical data analysis, with a particular emphasis on medical imaging and biomedical decision support. The central question driving my work is: *How can we design compact, reliable, and interpretable AI systems that remain dependable when real-world data are noisy, incomplete, corrupted, weakly supervised, or adversarially manipulated?*

This research direction aligns strongly with the Information & Computer Science areas advertised by King Fahd University of Petroleum & Minerals (KFUPM), especially **information security**, **secure computing**, **software engineering**, **high-performance computing**, **bioinformatics**, and **information visualization**. My work connects these areas through a unified agenda: developing secure and trustworthy learning systems that can support robust decision-making in biomedical and computational environments. As a faculty member, I aim to contribute to KFUPM by building student-centered research activities in trustworthy AI, secure learning systems, robust data analysis, and biomedical AI, while supporting undergraduate teaching and project supervision in the ICS Department.

My scholarly record currently includes **26 peer-reviewed or accepted scholarly works** across journal articles, international conference papers, and a book chapter, along with one manuscript currently under review at NeurIPS 2026. During my Ph.D., I have developed a focused research portfolio in robust quantum machine learning and trustworthy medical AI, including accepted or published work in **MICCAI**, **AAAI**, and the **Journal of Imaging**. These works form the foundation of my future research agenda in secure, robust, and efficient AI systems.

Current Research Contributions

1. Robust Quantum Machine Learning under Noisy Supervision

One of my major research contributions is the development of quantum-enhanced models for medical image classification under noisy labels. In my MICCAI 2025 work, **NQNN: Noise-aware Quantum Neural Networks for Medical Image Classification**, I addressed the problem of label noise in multiclass medical image benchmarks. Medical datasets often contain annotation errors due to expert disagreement, ambiguous pathology, or limited labeling resources. To address this challenge, I designed a noise-aware quantum neural network that integrates **Fourier attenuation**, **uncertainty-based sample reweighting**, and **adaptive pooling** within a variational quantum circuit framework. This work showed how quantum neural representations can be coupled with noise-aware learning mechanisms to improve robustness under 10%–50% label corruption.

This contribution is directly connected to secure and trustworthy computing because data corruption, annotation uncertainty, and unreliable supervision are common sources of failure in deployed AI systems. The broader lesson from this work is that robustness should not be treated only as an after-training evaluation metric; it should be integrated into the architecture, representation space, and training objective of the model.

2. Robust Medical AI under Corrupted Inputs and Weak Supervision

My AAAI 2026 work, **QAPNet: A Quantum-Attentive Patchwise Network for Robust Medical Image Classification under Noisy Inputs**, extends this research direction from noisy labels to corrupted imaging inputs and weak supervision. In real clinical settings, medical images may be degraded by acquisition noise, compression artifacts, missing information, scanner variability, and other forms of input corruption. QAPNet addresses this challenge through a patchwise multiple-

instance learning framework that combines sliding-window patch extraction, compact quantum patch encoding, additive-attention aggregation, contrastive alignment, and prototype regularization.

The research significance of this work is twofold. First, it treats robustness as a representation-learning problem rather than a simple preprocessing issue. Second, it provides an interpretable pathway for identifying which image regions contribute most strongly to the final prediction. This creates a natural connection to **information visualization** and clinical interpretability, both of which are relevant to KFUPM’s advertised areas. In future work, I plan to extend this direction by developing visual analytics tools that allow students and researchers to inspect latent-space organization, attention maps, prototype movement, and robustness behavior under different corruption settings.

3. Secure and Adversarially Robust AI against Backdoor Poisoning

My current under-review work, **HQADNet: Contrastive Prototype-Supervised Quantum Representation Learning for Robust Medical Image Classification against Backdoor Poisoning**, extends my research program toward security-oriented AI. Backdoor poisoning is a serious threat in which a model behaves normally on clean samples but produces attacker-specified outputs when a hidden trigger is present. This threat is particularly concerning in medical and biomedical AI, where training pipelines may involve third-party data, pretrained models, or distributed data collection.

HQADNet addresses this problem at the representation level. The framework uses a frozen classical encoder, PCA-based compact embedding, EM-refined soft prototypes, trust-aware training, and an 8-qubit tree-structured quantum encoder. The core idea is to prevent the model from learning shortcut trigger features by aligning quantum embeddings with class-consistent prototype structure. This research connects directly to **information security**, **secure machine learning**, and **secure computing**. It also provides a strong foundation for student projects in adversarial robustness, backdoor defense, AI security evaluation, and trustworthy representation learning.

4. Systems-Level and Trainability Support for Hybrid Quantum–Classical Learning

In addition to model development, I have contributed to systems-oriented research on hybrid quantum–classical learning. My Journal of Imaging work on **barren plateau mitigation** benchmarks initialization and optimization strategies for quantum neural networks across standard and medical image datasets. Another Journal of Imaging paper provides a systematic review of **systems-level support for hybrid quantum–classical learning**, with a medical imaging translation lens. Together, these works address an important question: how can quantum-enhanced learning methods become reproducible, trainable, and computationally practical?

This line of work aligns with **high-performance computing**, **software engineering**, and **AI systems and reproducibility**. Near-term quantum learning requires careful integration of simulation, GPU acceleration, circuit design, optimization, data movement, and software tooling. At KFUPM, this direction can support projects that combine AI systems, reproducible software pipelines, and emerging computing technologies.

Future Research Agenda at KFUPM

My future research agenda is designed to be feasible for a teaching-focused ICS appointment while also creating opportunities for undergraduate research, capstone projects, and interdisciplinary collaboration. I plan to develop the following connected research themes.

Theme 1: Secure and Trustworthy Machine Learning for Critical Data

I plan to build secure learning methods that are robust to data poisoning, backdoor triggers, noisy labels, distribution shifts, and unreliable supervision. This theme will involve designing evaluation protocols, representation-level defenses, uncertainty-aware training objectives, and diagnostic tools for detecting model failure. It aligns directly with KFUPM’s areas of information security, secure computing, data security, and secure software development.

Potential student projects include:

- benchmarking AI models under poisoned, corrupted, or shifted data;
- developing secure learning pipelines for image, text, and tabular data;
- building lightweight defense methods against backdoor and adversarial attacks;
- designing dashboards for visualizing model uncertainty and robustness failures.

Theme 2: Biomedical and Clinical AI with Robust Representation Learning

My second research theme focuses on biomedical AI and medical image analysis. I will continue developing models for robust image classification, weakly supervised diagnosis, noisy-label learning, and multimodal biomedical data analysis. This theme connects with KFUPM’s interest in **bioinformatics** and supports collaboration with health, biomedical, and data science researchers.

A key goal is to design models that are not only accurate, but also interpretable, uncertainty-aware, and reproducible. This includes prototype-based learning, attention-guided representation analysis, calibration methods, and visual explanations. These components are valuable for teaching students how to move beyond black-box AI and develop responsible systems for high-impact domains.

Theme 3: Efficient Hybrid Quantum–Classical and High-Performance AI Systems

My third research theme concerns computational efficiency and system-level support for emerging AI models. Hybrid quantum–classical learning provides a useful platform for studying compact representation learning, trainability, optimization, and simulation efficiency. Although near-term quantum hardware is still developing, quantum-inspired and quantum-simulated methods can already motivate research in efficient AI, high-performance simulation, and reproducible software systems.

At KFUPM, this theme can support student projects involving GPU-based model training, quantum circuit simulation, performance benchmarking, software engineering for reproducible AI pipelines, and high-performance computing for biomedical data analysis. The goal is not to overstate quantum advantage, but to study when and how compact quantum-enhanced representations can improve robustness, efficiency, or interpretability.

Theme 4: Information Visualization for Trustworthy AI

My fourth research theme is to develop visualization tools for understanding model behavior. Robustness and security problems are often difficult to diagnose because failures occur in hidden representation spaces. I plan to develop visual analytics methods for prototype movement, attention maps, uncertainty distributions, corrupted-input behavior, and attack/defense outcomes. This line of work connects strongly to KFUPM’s advertised area of **information visualization** and can also support undergraduate projects because it combines machine learning, software development, user-centered design, and data analysis.

Student Involvement and Research Mentoring

My research philosophy is closely connected to student mentoring. Before beginning my Ph.D., I held full-time academic appointments as Lecturer, Senior Lecturer, and Assistant Professor, and I supervised substantial undergraduate project and thesis work. I plan to continue this student-centered approach at KFUPM by creating structured research pathways for Bachelor-level students in the ICS Department.

For early-stage students, I will offer projects focused on reproducible experiments, dataset preparation, visualization, and baseline implementation. For advanced students, I will supervise projects involving secure AI evaluation, robust machine learning, medical image analysis, high-performance AI experiments, and software tools for trustworthy computing. This approach will help students develop both theoretical understanding and practical engineering skills.

I also aim to involve students in publishable research by guiding them through problem formulation, literature review, implementation, experimentation, writing, and presentation. This is consistent with the KFUPM Lecturer role, which emphasizes quality education, student interaction, and contribution to the academic mission of the department.

Collaboration and Long-Term Impact

My long-term goal is to develop a research program that contributes to trustworthy computing for safety-critical applications. This includes secure AI, robust biomedical data analysis, high-performance AI systems, and interpretable model development. At KFUPM, I would seek collaboration with colleagues in information security, software engineering, high-performance computing, data science, and biomedical computing. I also plan to contribute to externally visible research outputs through student projects, conference submissions, journal publications, and reproducible open-source implementations.

My prior experience as principal investigator on funded research projects in machine learning and cyberbullying detection, together with my current Ph.D. research productivity, provides a strong foundation for developing future collaborative research proposals. I am particularly interested in projects that combine secure AI, biomedical data, and scalable computing infrastructure.

Conclusion

In summary, my research program is built around a coherent objective: designing **secure, robust, interpretable, and computationally efficient AI systems** for complex and high-impact data environments. My work in noisy-label learning, corrupted-input robustness, backdoor poisoning defense, quantum machine learning, and systems-level support for hybrid quantum–classical AI aligns well with the KFUPM ICS areas of information security, secure computing, software engineering, high-performance computing, bioinformatics, and information visualization. As a Lecturer in the Information & Computer Science Department, I would contribute through high-quality undergraduate teaching, active student mentoring, and a focused research agenda that connects trustworthy AI with secure and data-intensive computing.